SABARAGAMUWA UNIVERSITY OF SRI LANKA
FACULTY OF COMPUTING
DEPARTMENT OF SOFTWARE ENGINEERING
BSc HONOURS DEGREE PROGRAMME IN SOFTWARE ENGINEERING

SE 8101 Research Project

# Research Proposal

| | | | |
|---|---|---|---|
| Student Name | S.Shanthosh | Student Index Number | 20APSE4882 |
| Research Topic | An Empirical Evaluation of Deep Learning Models for Enhanced and Scalable Bug Detection and Classification in Large-Scale Software Systems | | |
| Supervisor(s) | Mrs.WMLS.Abeythunga | | |

**Agreement**

I …… S.Shanthosh…… ……, willing to submit the following Proposal format and I am confirming that this topic is a novel research and I am aware of the consequences occurs in the case of plagiarism found.

…….  16-01-2026
Signature  Date

Background

## The Growing Challenge of Software Defects in Complex Systems

The modern software landscape is defined by its scale and complexity, characterized by massive codebases, continuous deployment schedules, and distributed architectures (e.g., microservices, cloud-native applications). In this environment[2], software defects (bugs) are not just an inconvenience; they are a major cost driver and a direct threat to system reliability and security. Studies show that the cost of fixing a bug increases exponentially the later it is discovered in the Software Development Lifecycle (SDLC)[4],[6]. Traditional bug detection methods, such as manual code review and rule-based static analysis, are proving increasingly inadequate for these large-scale systems.

## Limitations of Traditional Detection Methods

Conventional static analysis tools rely on heuristic rule sets[7],[9], which often result in a high False-Positive Rate (FP). This forces developers to waste time investigating benign code, reducing confidence in the tools and lengthening the debugging cycle. Furthermore, these tools are generally [12]ineffective at detecting complex, context-dependent runtime errors, memory leaks, or race conditions—the very types of bugs prevalent in large, concurrent systems. Dynamic analysis is resource-intensive and often cannot cover all execution paths, leaving critical vulnerabilities undiscovered[16].

Background

**The Rise of Artificial Intelligence in Software Quality Assurance**

To address these limitations, the software engineering community has turned to Artificial Intelligence (AI), specifically Machine Learning (ML) and Deep Learning (DL), to develop intelligent, self-learning debugging systems[14]. AI models can analyze vast amounts of data—including source code features, process metrics[22], and natural language from bug reports—to learn complex patterns indicative of defects[24]. This capability allows for predictive analytics, enabling the identification of defect-prone modules *before* code is executed, or providing automated classification of bug reports, significantly streamlining the bug triaging process in large teams..

**The Critical Research Gap in Scalable AI Bug Detection**

While numerous ML models (e.g., Random Forest, Naïve Bayes) have been successfully applied to smaller, benchmark datasets like the PROMISE repository[14], applying and optimizing these techniques for *large-scale, real-world* systems presents significant challenges. The primary gap lies in: **(1)** Developing DL architectures [19],[21],[23](like CNN-LSTM or Transformer-based models) that can effectively model the semantic and structural complexities of massive codebases. **(2)** Overcoming data-related challenges, specifically the high class imbalance (defective code modules are rare) and the need for **cross-project generalization**, which is essential for models to be practically adopted by the industry. This study directly aims to bridge this gap by focusing on highly accurate and scalable DL techniques for both bug prediction and automated bug report classification.

Literature Review

## Introduction to Software Defect Prediction and Classification

The field of AI-driven bug detection primarily bifurcates into Defect Prediction and Bug Report Classification. Defect prediction involves predicting whether a code module[24],[25] (e.g., a file or function) is fault-prone using metrics like Cyclomatic Complexity and Halstead's measures. Bug Report Classification involves analyzing the textual summary and description of an issue (e.g., from Bugzilla) to categorize it by type (e.g., functional, security, performance) or assign a priority.

## Traditional Machine Learning Techniques

confidence in the tools and lengthening the debugging cycle. Furthermore, these tools are generally ineffective at detecting complex, context-dependent runtime errors, memory leaks, or race conditions[30],[6]—the very types of bugs prevalent in large, concurrent systems. Dynamic analysis is resource-intensive and often cannot cover all execution paths, leaving critical vulnerabilities [8] undiscovered.

Literature Review

# Deep Learning Architectures for Code and Text Analysis

Deep Learning (DL) models are increasingly employed for their ability to automatically learn complex features from raw data[14],[15],[16]:

- Code-Based Models: Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM), are used to analyze code sequences to identify spatial and sequential patterns that signal defects. Hybrid models, such as CNN-LSTM, have shown superior performance by capturing both local and long-range dependencies in source code metrics.

- Text-Based Models: Transformer-based models like [9],[10] BERT (Bidirectional Encoder Representations from Transformers) and GPT-3 are now state-of-the-art for analyzing natural language in bug reports. They generate contextual embeddings, allowing them to understand the semantic pattern within bug descriptions for automated classification, triaging[15],[16], and even automated fix generation.

## Empirical Benchmarks and Datasets

Empirical evaluation of bug detection models relies on publicly available, real-world datasets [35],[36]:

- **Defects4J:** A widely-used database providing reproducible faults for Java programs, essential for evaluating fault localization and automated program repair techniques.
- **PROMISE Repository:** Contains datasets (e.g., JM1, KC1, PC1) derived from NASA and other projects, primarily used for traditional defect prediction based on software metrics.
- **Bugzilla Issue Trackers and GitBugs:** These repositories provide large-scale collections of bug reports with comprehensive metadata (summary, description, status, priority), which are crucial for classification tasks and NLP-based approaches.

## Performance Metrics and Challenges

Model performance is typically assessed using standard classification metrics, as required by this study:

- Accuracy: Overall correctness: (TP + TN) / (Total).
- Precision (P): Correctness of positive predictions (minimizing false positives).
- Recall (R): Ability to find all positive instances (minimizing false negatives).

F1-Score: The harmonic mean of Precision and Recall, a robust measure for imbalanced datasets.

A major challenge is the class imbalance problem[18],[14], where the number of non-defective modules vastly outweighs defective ones. Techniques [7] like SMOTE (Synthetic Minority Oversampling Technique) are critical for re-balancing datasets and preventing the model from achieving high accuracy while failing to detect the minority (buggy) class.

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Literature Review

| Reference | Objective | Methodology | Limitations |
|---|---|---|---|
| **[41]** | To predict defect-prone software modules using deep learning on code metrics. | Deep Neural Networks trained on PROMISE datasets; use of software complexity metrics (CK, Halstead); comparison with traditional ML models (RF, SVM). | Limited semantic understanding of source code; relies heavily on handcrafted metrics; reduced performance in cross-project prediction. |
| **[32]** | To improve bug detection by modeling semantic and structural code relationships. | Graph Neural Networks (GNNs) applied to Abstract Syntax Trees (ASTs) and code graphs; node embeddings used for defect classification. | High computational cost; complex model architecture; requires large labeled datasets for stable performance. |
| **[14]** | To evaluate the effectiveness of CNN-based models for software defect prediction. | Convolutional Neural Networks applied to tokenized source code and metric-based inputs; empirical evaluation on benchmark datasets. | CNNs capture local patterns but struggle with long-range dependencies; limited explainability of predictions. |
| **[22]** | To automate bug report classification using contextual language understanding. | BERT-based transformer models fine-tuned on Bugzilla datasets; NLP preprocessing and supervised classification. | Requires large computational resources; performance sensitive to domain-specific vocabulary; limited integration with source code features. |

Gaps in Existing Research – Detailed Study

While existing research has proven the feasibility of AI in bug detection, three significant gaps remain, particularly for **large-scale industrial adoption**:

**1.Lack of Semantic-Structure Fusion in Code Defect Prediction**: Current models often treat code as purely a sequence (NLP-style) or rely only on high-level metrics (traditional ML). There is a critical gap in approaches that effectively fuse semantic features (from token sequences) and structural features (from Abstract Syntax Trees/ASTs) to create a comprehensive representation vector for prediction, which is essential for improving F1-score and reducing false negatives in complex code.

**2.Poor Cross-Project Generalization:** Most models achieve high performance on the specific dataset they are trained on (within-project validation). They often fail when tested on projects with different architectures, languages, or coding styles (cross-project validation). This lack of generalizability is a key barrier to industrial deployment. The research needs to explore domain adaptation or transfer learning techniques to build models that perform reliably across diverse, large-scale systems.

**3.Limited Explainability in Deep Learning Models:** DL models, despite their high accuracy, often operate as "black boxes." In the safety-critical domain of software quality, developers require not just a prediction ("this module has a bug"), but also an explanation ("the bug is likely due to an inconsistency between the name and its implementation at line X"). The current gap is the integration of high-accuracy DL models with interpretable AI (XAI) techniques to provide actionable, human-readable rationales for the predicted defects

## 4.Scarce and Inconsistent Industrial-Scale Labeled Data :

The performance of sophisticated deep learning models is inherently constrained by the quantity and quality of their training data. Current academic research often relies on publicly available datasets (like those from GitHub or historical defect reports), which are frequently noisy, inconsistently labeled, and do not accurately reflect the complexity and scale of proprietary industrial codebases. The critical gap is twofold: first, the scarcity of large, high-quality, and reliably labeled datasets that link specific code patterns to confirmed defects *and* their corresponding fixes. Second, the inconsistency in labeling standards across different organizations and projects makes it difficult to aggregate data and train a universally robust model. Industrial deployment requires a framework to efficiently and ethically curate vast amounts of proprietary defect data with high labeling fidelity to sustain the performance of AI models over the project's lifecycle, addressing issues like concept drift as the codebase evolves.

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Problem Statement and Research Questions

## Problem Statement

Current AI-driven bug detection techniques, while promising, face significant limitations in terms of scalability, generalizability, and interpretability when applied to large-scale, complex software systems. This leads to continued reliance on time-consuming manual debugging, high false-positive rates, and delayed discovery of critical defects, consequently increasing software development costs and compromising system reliability.

## Research Questions

**RQ1:**
How effectively do traditional machine learning models (Random Forest, Naïve Bayes) predict software defects compared to deep learning models in large-scale software systems?

**RQ2:**
How do deep learning models (CNN-LSTM, BERT-based models) improve bug detection and classification performance on large and imbalanced software defect datasets?

**RQ3:**
What is the impact of dataset characteristics (Defects4J and PROMISE) on the performance, scalability, and generalization of ML and DL-based defect prediction models?

**RQ4:**
Which model achieves the best balance between accuracy, precision, recall, and F1-score for scalable software defect detection and classification?

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## Problem Statement and Research Questions

**Main Problem**
AI-based bug detection struggles with scalability, generalizability, and interpretability in large software systems, causing high false positives, delayed defect discovery, and increased development costs

**Main Objective**
To empirically evaluate and compare machine learning and deep learning models for accurate and scalable software bug detection and classification using benchmark datasets

**RQ1:**
How effectively do traditional machine learning models (Random Forest, Naïve Bayes) predict software defects compared to deep learning models in large-scale software systems?

**O1:**
To implement traditional machine learning models (Random Forest and Naïve Bayes) for software defect prediction.

**RQ2:**
How do deep learning models (CNN-LSTM, BERT-based models) improve bug detection and classification performance on large and imbalanced software defect datasets?

**O2:**
To design and apply deep learning models (CNN-LSTM and BERT-based models) for enhanced bug detection and classification.?

**RQ3:**
What is the impact of dataset characteristics (Defects4J and PROMISE) on the performance, scalability, and generalization of ML and DL-based defect prediction models?

**O3:**
To evaluate model performance using Defects4J and PROMISE datasets under imbalanced data conditions.

**RQ4:**
Which model achieves the best balance between accuracy, precision, recall, and F1-score for scalable software defect detection and classification?

**O4:**
To compare ML and DL models using standard evaluation metrics (Accuracy, Precision, Recall, F1-score) and identify the most effective approach.

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| Research Objectives and Goals |
| --- |
| **Primary Goal (Aim)**<br>To develop, implement, and empirically validate a highly accurate, robust, and scalable AI-driven framework for the automatic detection and classification of software bugs in complex, large-scale software systems using cutting-edge Deep Learning techniques. |

| Research Objectives and Goals |
| --- |
| **Specific Objectives** To achieve this primary goal, the following specific objectives have been formulated, each corresponding to the research questions:<br><br>• Comparative Analysis: Systematically compare the performance of four established ML/DL algorithms (e.g., Random Forest, Naïve Bayes, CNN-LSTM, and BERT-based classifier) on benchmark software defect datasets (Defects4J, PROMISE, and GitBugs).<br>• Model Development and Optimization: Design and implement a novel multi-modal DL model that fuses features extracted from source code metrics, Abstract Syntax Trees (ASTs), and natural language bug report text to enhance predictive power.<br>• Data Preprocessing and Balancing: Investigate and apply advanced data preprocessing techniques, including SMOTE and feature dimensionality reduction (e.g., PCA), to mitigate the class imbalance problem inherent in software defect data.<br>• Empirical Validation: Rigorously evaluate the developed model against the current state-of-the-art using the target performance metrics (Accuracy, Precision, Recall, and F1-Score), ensuring superior performance, especially in terms of F1-Score on the minority (defective) class.<br>• SMOTE is applied as a baseline oversampling technique to address class imbalance commonly observed in software defect datasets, ensuring fair model training and evaluation.<br>The study also acknowledges more advanced imbalance-handling approaches, positioning SMOTE as a controlled baseline |

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Proposed Research Method

**Research Design**

This study will adopt a Quantitative, Comparative, and Empirical Research Design. It will be primarily experimental, focused on developing and comparing the classification performance of various machine learning models using real-world software defect data.

**Data Collection Strategy (Quantitative)**

The research will utilize publicly available, established datasets to ensure reproducibility and provide a clear basis for comparison with existing literature.

- Datasets: Defects4J (code-based defects), a selection of projects from the PROMISE Repository (metric-based defects), and the GitBugs/Bugzilla dataset (textual bug reports for classification).
- Feature Extraction: Features will be extracted to support two model types:
    - Metric-based: Software complexity metrics (e.g., McCabe's Cyclomatic Complexity, Halstead's metrics).

Sequence/Semantic-based: Code tokens, Abstract Syntax Tree (AST) information, and word embeddings from bug report text.

**Data Analysis Plan (Quantitative)**

The methodology will follow a standard ML pipeline:

1. Preprocessing: Clean and normalize the data. Crucially, address class imbalance using techniques like Borderline-SMOTE or SMOTETomek to synthesize minority class samples. Dimensionality reduction (e.g., PCA or Recursive Feature Elimination) will be used to optimize feature sets.
2. Model Training: Train both baseline models (RF, NB) and advanced DL models (CNN-LSTM, BERT) on the preprocessed training datasets. Hyper-parameter tuning will be performed using techniques like Random Search or Grid Search to maximize model performance.

3. Model Evaluation: Performance will be evaluated using 10-fold cross-validation and a separate hold-out test set. The primary evaluation focus will be the F1-Score, Precision, and Recall, which are critical indicators of real-world utility in imbalanced scenarios.

Generalization Test: A cross-project validation will be performed where a model trained on Project A is tested on the test set of Project B to assess the model's robustness and generalizability.

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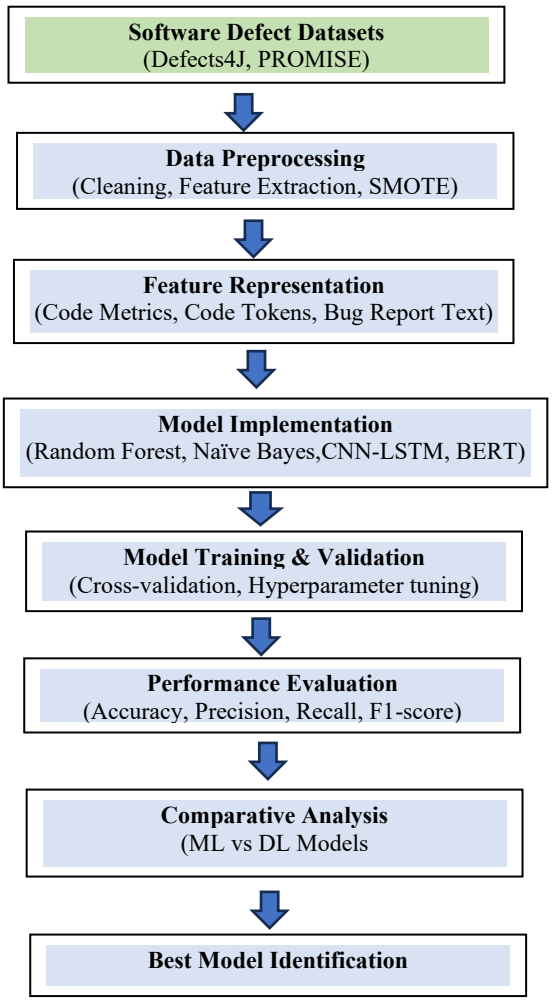
**Ethical Considerations and Validity**

The study relies entirely on public-domain, open-source software datasets (Defects4J, PROMISE, Bugzilla), ensuring that no proprietary or private developer data is used, adhering to ethical research standards.

Proposed Research Method

Validity will be ensured by using accepted benchmarking protocols, such as using metrics specific to software defect prediction (e.g., F1-Score) and employing cross-val



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**Data Sources and Usage**

This research utilizes two publicly available software defect datasets: Defects4J and PROMISE. Defects4J provides real-world Java program bugs with corresponding test cases, while PROMISE contains software metrics-based defect data. These datasets are used to train, validate, and evaluate machine learning and deep learning models for defect prediction and classification.

The datasets are preprocessed through cleaning, normalization, and handling of class imbalance. Features are extracted and fed into machine learning models (Random Forest, Naïve Bayes) and deep learning models (CNN-LSTM, BERT). The trained models are evaluated using standard performance metrics, and results are compared to identify the most effective approach.

Expected Outcome

**Primary Outcome: A Multi-Modal DL Bug Detection Framework**

The principal outcome of this research will be the development of an evidence-based, Multi-Modal Deep Learning Bug Detection Framework. This framework will integrate code features (structural and semantic) and bug report text features to provide a superior, scalable solution for bug detection and classification.

**Specific Deliverables**

1. A Validated DL Model: A trained and optimized Deep Learning model (e.g., fused CNN-LSTM or BERT-based) demonstrably achieving higher F1-scores on the minority (defective) class across multiple benchmark datasets than current state-of-the-art models.

2. Comparative Analysis Report: A comprehensive report detailing the comparative performance of traditional ML versus DL models, including analysis of feature importance, model complexity, and training overhead.

3. Practical Guidelines: A set of actionable guidelines for software organizations on the optimal feature sets, preprocessing techniques, and model selection criteria for deploying AI-driven bug detection solutions in their large-scale production environments.

4. Academic Publication: A research paper submitted to a reputable peer-reviewed journal or conference (e.g., IEEE/ACM) detailing the methodology and empirical results.

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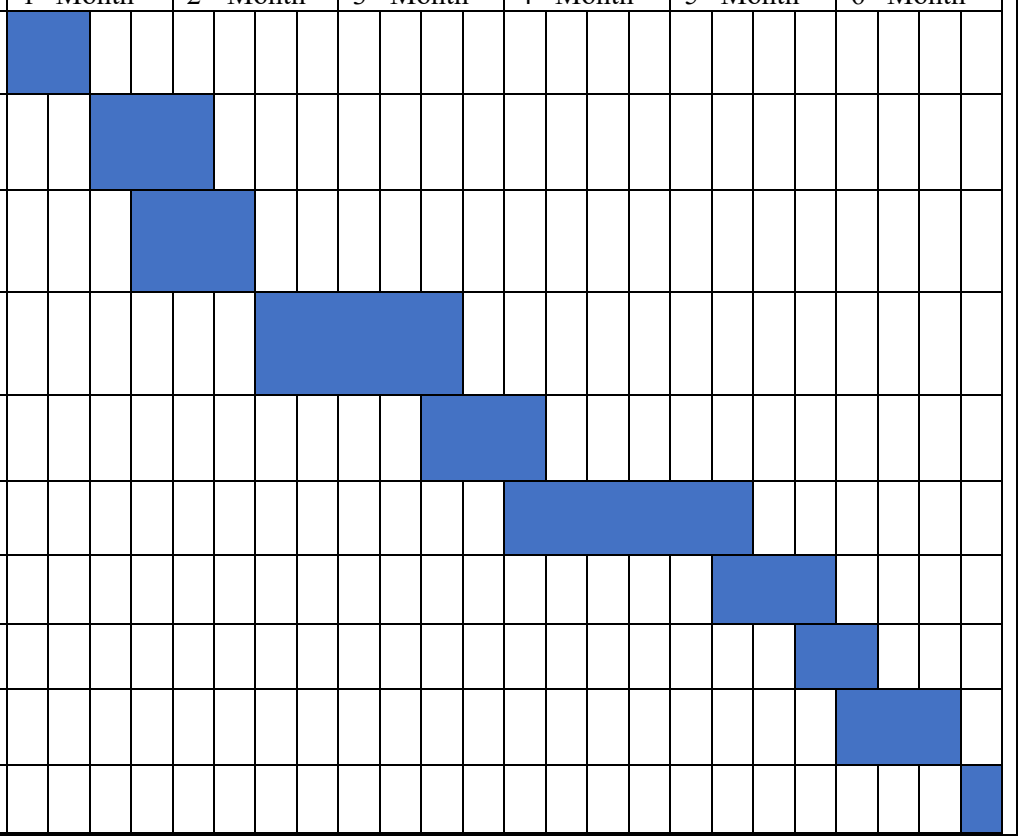
## Research Plan and Time Table

| Activity | Time Duration | | | | | |
|---|---|---|---|---|---|---|
| | 1st Month | 2nd Month | 3rd Month | 4th Month | 5th Month | 6th Month |
| Topic Selection | | | | | | |
| Literature Review | | | | | | |
| Preparing Research Proposal | | | | | | |
| Dataset Collection and Preparation | | | | | | |
| Model Selection | | | | | | |
| Implementation | | | | | | |
| Testing & Evaluation | | | | | | |
| Result Analysis | | | | | | |
| Report Writing | | | | | | |
| Final Submission | | | | | | |



## Proposed Budget

This research does not require direct financial expenditure because all tools, software frameworks, and programming environments used in the study are freely accessible. Python, scikit-learn, PyTorch, CodeBERT, CodeT5, and other required libraries are available as open-source packages. AI code samples will be generated using free-tier access or publicly available models, and datasets will be created from openly accessible coding problems such as HumanEval and MBPP. Code execution and experimentation will be carried out on standard computing machines already available, eliminating the need for additional hardware purchases. No paid datasets, commercial APIs, or licensed software tools are required, enabling the project to be completed at zero cost.

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**Research Novelty**

"Although software defect prediction has been widely studied, this research provides a unified empirical comparison of traditional machine learning and deep learning models using both code-level and bug-report-level datasets. The study further analyzes model generalization across heterogeneous datasets under imbalanced conditions, offering practical insights for model selection in large-scale software systems."

**Model Description**

- **Random Forest**

"Random Forest is selected due to its robustness, ability to handle high-dimensional data, and proven effectiveness in traditional defect prediction studies."

- **BERT**

"BERT is chosen for its strong contextual understanding of textual bug reports, enabling improved classification performance compared to traditional text models."

**Expected Outcomes**

- Identification of the most effective model for defect prediction and classification
- Comparative performance analysis of ML and DL approaches
- Insights into dataset impact on model generalization
- Practical guidance for selecting defect prediction models

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| Any Other Relevant Information |
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| N/A |